$$P = \frac{a}{i}\left[1 - \frac{1}{(1+i)^n}\right]$$

Now as per the definition of perpetual annuity as $n \to \infty$, we know that $\frac{1}{(1+i)^n} \to 0$ since $1+i > 1$.

Here $P = \frac{a}{i}[1-0]$

$$P = \frac{a}{i}$$

NOTE

In all the above formulae the period is of one year. Now if the payment is made more than once in a year then 'i' is replaced by $\frac{i}{k}$ and n is replaced by nk, where k is the number of payments in a year.

Example 7.1

A person pays ₹ 64,000 per annum for 12 years at the rate of 10% per year. Find the amount of an ordinary annuity $[(1.1)^{12}=3.3184]$.

Solution:

Here $a = 64{,}000$, $n = 12$ and $i = \frac{10}{100} = 0.1$

Amount of ordinary annuity $(A) = \frac{a}{i}[(1+i)^n - 1]$

$= \frac{64000}{0.1}[(1+0.1)^{12} - 1]$

$= 6{,}40{,}000[(1.1)^{12} - 1]$

$= 6{,}40{,}000[3.3184 - 1]$

$= 6{,}40{,}000\ [2.3184]$

$= 64 \times 23184$

$\therefore$ A = ₹ 14,83,776

Example 7.2

What amount should be deposited annually in an ordinary annuity scheme, so that after 16 years, a person receives ₹1,67,160 if the interest rate is 15% $[(1.15)^{16}=9.358]$.

Solution:

Here $A = 1{,}67{,}160$, $n = 16$ $i = \frac{15}{100} = 0.15$ $a = ?$

To find : a

Now $A = \frac{a}{i}[(1+i)^n - 1]$

$1{,}67{,}160 = \frac{a}{0.15}[(1+0.15)^{16} - 1]$

$= \frac{a}{0.15}[(1.15)^{16} - 1]$

$\Rightarrow a = \frac{1{,}67{,}160 \times 0.15}{(1.15)^{16} - 1}$

$= \frac{1{,}67{,}160 \times 0.15}{9.358 - 1} = \frac{1{,}67{,}160 \times 0.15}{8.358} = 3{,}000$

Therefore a = ₹ 3,000

Example 7.3

The age of the girl is 2 years. Her father wants to get ₹20,00,000 when his ward becomes 22 years. He opens an account with a bank at 10% rate of compound interest. What amount should he deposit at the end of every month in this recurring account? $[(1.0083)^{240}=6.194]$.

Solution:

Here $A = 20{,}00{,}000$; $i = \frac{10}{100} = 0.1$ $n = 20$ and $k = 12$

$A = \frac{a}{i/k}[(1+\frac{i}{k})^{nk} - 1]$

$20{,}00{,}000 = \frac{a}{\frac{0.1}{12}}\left[\left(1 + \frac{0.1}{12}\right)^{20 \times 12} - 1\right]$

$= \frac{12a}{0.1}\left[\left(1 + \frac{0.1}{12}\right)^{240} - 1\right]$

$= 120a\left[\left(\frac{12.1}{12}\right)^{240} - 1\right]$

$= 120a\ [(1.0083)^{240} - 1]$

$= 120a\ [6.194 - 1]$

$= 120a\ (5.194)$

$\Rightarrow a = \frac{20.00{,}000}{120 \times 5.194} = 3208.83$

$\therefore a \approx 3{,}209$

₹3,209 is to be deposited at the end of every month.

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